DAVID ALEXANDER GARCIA MELGAR

SENIOR SOFTWARE ENGINEER | GPS FLEET PLATFORMS, BACKEND & FULL-STACK

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PROFESSIONAL SUMMARY

Senior Software Engineer with 20+ years of hands-on experience designing, developing, optimizing, and supporting production enterprise applications, real-time GPS fleet tracking platforms, telecommunications systems, and high-volume relational databases. Strong hands-on C# and backend integration experience, with introductory Ruby experience from a small project. Experienced in scalable APIs, Angular and TypeScript, Node.js, SQL/MySQL, Oracle PL/SQL, system integration, and performance-critical applications. Delivered fleet monitoring and operational systems serving large distributed field teams, with a strong focus on reliability, maintainable architecture, and solving complex production issues.

TECHNICAL SKILLS

Languages: C# (.NET), JavaScript, TypeScript, SQL, PL/SQL, Python, Java, PHP, C++, Kotlin; Ruby (small project exposure)

Frontend & Mobile: Angular, React, Next.js, React Native, HTML, CSS, Android

Backend & Integration: C#/.NET, Node.js, Express, REST API design, SOAP, WS-Security, object-oriented programming, microservices, system integration

Databases: MySQL, Oracle 11g/12c/19c/21c, PostgreSQL, SQL Server, MongoDB, BigQuery; query tuning, indexing, large datasets

Infrastructure & Delivery: Docker, Kubernetes, Linux, Nginx, Git, Redis, Kafka, CI/CD concepts, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Sternemedia S. de R.L. | Apr 2014 – Present

- Developed and supported production software for GPS fleet tracking, telecom field operations, enterprise reporting, and web-based services, including hands-on backend and frontend implementation.
- Worked with engineering teams of up to 20 professionals on application delivery, technical design, complex troubleshooting, code review, and production issues.
- Built and supported operational systems monitoring more than 2,200 telecommunications sites, integrating live operational data, alerts, reporting, and distributed field-team workflows.
- Engineered SQL-intensive applications with Oracle PL/SQL packages, stored procedures, indexes, query optimization, and integrations with Node.js and Angular/React interfaces; also worked with MySQL.
- Implemented cross-border government interoperability using C#, SOAP, and Oracle, reducing integration errors from approximately 48% to 2.5% in a workflow processing about 6,000 exchanges.
- Developed enterprise web solutions with Angular, TypeScript, React, Node.js, and Python; deployed and troubleshot containerized workloads using Docker, Kubernetes, and Linux.
- Delivered production solutions used by Tigo (Millicom), Claro, and Digicel; associated supplier evaluations reached 92% in 2023 and 100% in 2024.

SELECTED PROJECTS

- Telecom GPS & Fleet Monitoring (Eagleview): Developed real-time GPS tracking dashboards, vehicle location and movement reporting, fatigue analytics, alerts, access authorization workflows, and performance-optimized Oracle reporting for distributed field operations.
- Government Digital Services: Built online renewal and international interoperability services, including secure APIs, C#/SOAP interfaces, and Oracle-backed integration with legacy production systems.
- Field Reporting & Automation: Designed ticketing and evidence-capture workflows for Telegram and web clients, using Node.js, Oracle packages, role-based access, reporting templates, and AI-assisted review.
- Cloud-Native Deployment: Containerized Node.js applications and operated Kubernetes deployments with database persistence, internal service networking, troubleshooting, and production diagnostics.

CERTIFICATIONS & EDUCATION

- Oracle SQL & PL/SQL training/certification (2004) | Scrum Master | Microsoft Azure Fundamentals
- Systems Engineering studies — Honduras 2007
- University of British Columbia — Software developer bootcamp 2025